

Analysis of multi-light image collections for improved visualization of historic textiles



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Abstract: This study investigates the use of Reflectance Transformation Imaging (RTI) for the analysis of historic textiles, employing the Relight software to generate relightable visualizations from multi-light image collections (MLICs). Various algorithms within Relight were compared in terms of their ability to accurately reconstruct scene appearance. Visual inspections and quantitative evaluations using CIELAB color difference formula was performed. However, challenges such as desaturation in the reconstructed images highlight limitations that require further investigation.

INTRODUCTION

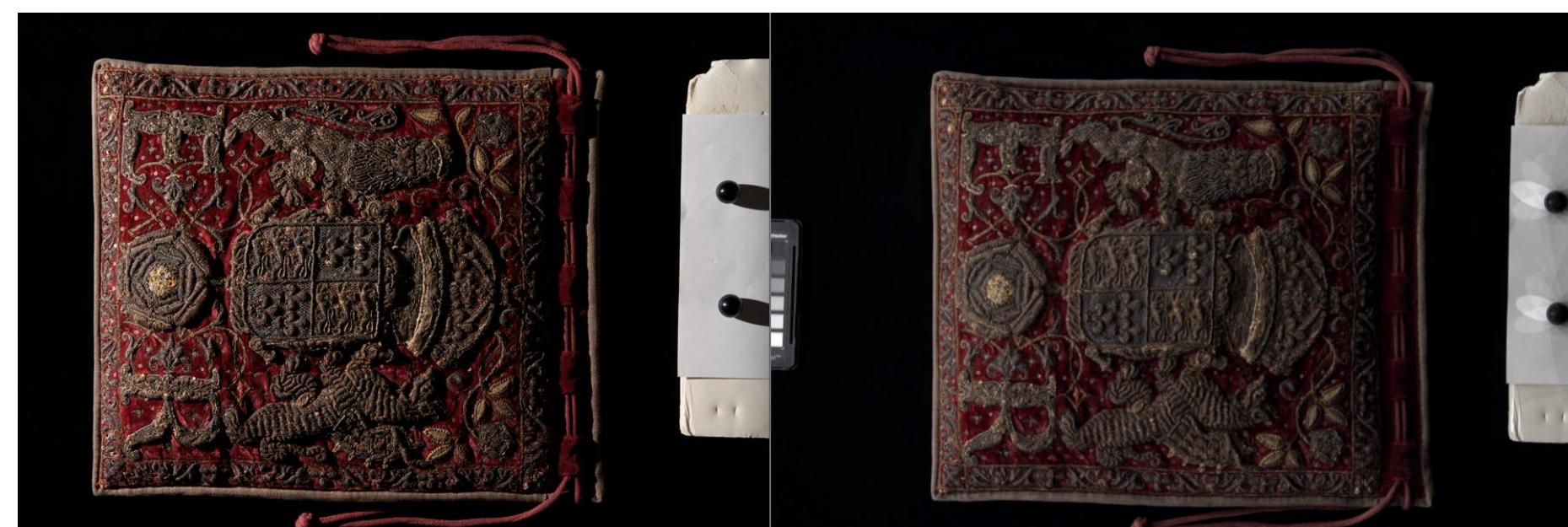
RTI (Reflectance Transformation Imaging):

Using an input image set, the scene's appearance is encoded through a set of basis coefficients, allowing for the dynamic relighting of the scene from unrecorded light positions.

Project Goal: This project explores the use of Relight software [1][2] to process these multi-light images and create visualizations that can be interactively relit.

ALGORITHMS IN RELIGHT

- **LPTM and PTM:** Polynomial Texture Mapping (18 or 9 planes).
- **HSB:** Hemispherical Harmonics (12 or 27 planes).
- **RBF:** Radial Basis Function interpolation with a variable number of planes.
- **YRBF:** An enhanced version of RBF that processes colors separately in the JPEG color space, improving chroma fidelity.
- **BLN and YBLN:** These algorithms use bilinear interpolation instead of RBF.



On the left is the input image, from the image set, and on the right is the reconstruction using the YBLN algorithm at the same light angle

RESULTS AND LIMITATIONS

Visual Inspection – First, the reconstructed RTI images were compared to input images at similar light angles to assess accuracy of algorithms.

Quantitative Analysis – average CIELAB2000 colour difference between original and reconstructed images were computed for 5 main algorithms across 6 image sets.

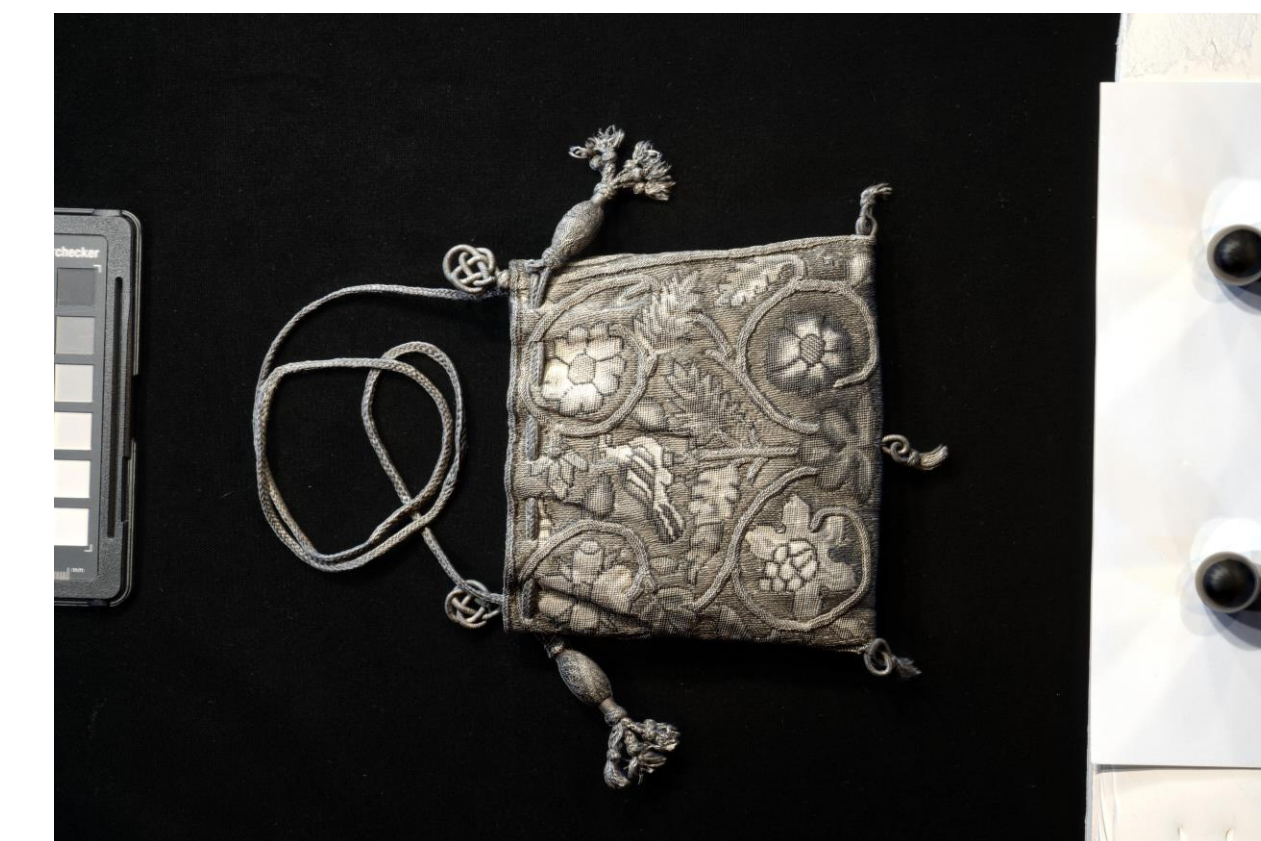
Imageset	Bilinear	LPTM	PTM	RBF	YRBF
T.40 1986 Back	2.8628	3.6128	3.5172	2.486	7.2339
T.40 1986 Front	4.8083	5.054	4.9605	4.2048	6.0964
T.52 1954 Back	5.4108	6.7249	6.5948	4.9935	5.2157
T.52 1954 Front	5.5653	7.1047	6.914	4.9468	5.1407
T.52A 1954 Back	4.5597	6.2149	6.1397	4.0749	4.0478
T.52A 1954 Front	4.3685	5.7805	5.6806	4.0464	4.0199

Average CIELAB color difference for each image set, calculated by first averaging the pixel-by-pixel color differences between the input and output images for each image, and then averaging those results across the entire image set.

Desaturation problem – A significant challenge was the desaturation of colors in the RTI images compared to the originals. This issue persisted despite various troubleshooting efforts, such as excluding shadow-heavy images, and testing the software on different operating systems.

FUTURE DIRECTIONS

Addressing Desaturation – Further research is needed to determine the cause of the desaturation – whether it's due to the software, the datasets, or the RTI creation process.



Example of a heavily desaturated reconstruction. (RBF algorithm, T.52 Front image)

Broader Testing – The software should be tested on a wider range of datasets to identify potential improvements.

Software Enhancements – Improvements could include developing a more streamlined workflow and adding features like the ability to take snapshots of the RTI visualizations within Relight.

REFERENCES

[1] Github page for Relight. <https://github.com/cnr-isti-vclab/relight>

[2] Relight project site: <https://vcg.isti.cnr.it/vcgtools/relight/>